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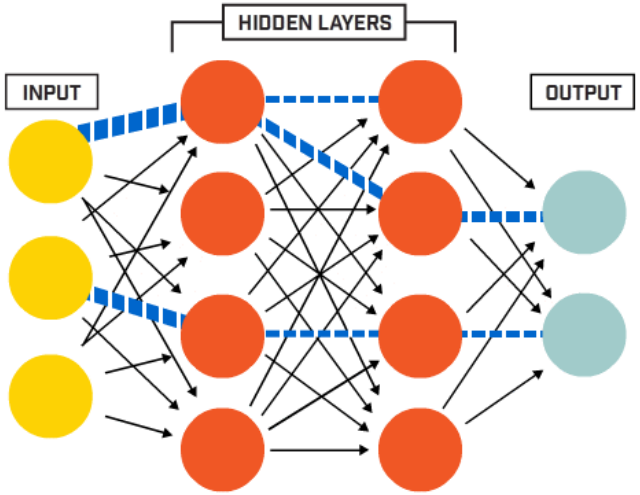
DEEP LEARNING BASICS



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Unit -4

7.1 Biological Motivation

7.1.1 Structure and Function of Biological Neurons

7.1.2 How Neurons Communicate: Synapses

7.2 Neural Network Representation

7.2.1 Basic Structure of an Artificial Neuron

7.2.2 Activation Functions

7.2.3 Layers and Connections

7.3 Problems for Neural Network Learning

7.3.1 Classification Tasks

7.3.2 Regression Problems

7.3.3 Pattern Recognition

7.3.4 Time-Series Prediction

7.4 Perceptron and Multi-layer Networks

7.4.1 Structure of Perceptron

7.4.2 Learning algorithm for Perceptron

7.4.3 Single-layer and Multi-layer Perceptron

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- **Neural Networks**
  - Brain – complex, non linear and parallel computer - capability of organize its structural constituents called neurons so as to perform tasks like pattern recognition, perception, motor control etc
  - A Neural network is a massively parallel distributed processor made up of simple processing units which has a natural propensity for storing experiential knowledge and making it available for use. It resembles human brain in two aspects
    - Knowledge is acquired by the network from its environment through a learning process
    - Interneuron connection strengths known as synaptic weights are used to store the acquired knowledge
  - Procedure to perform the learning process is called learning algorithm

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- **Benefits of Neural Networks**
  - Neural Networks achieve power through massively Parallel and Distributed Structure
  - Generalization – refers to produce reasonable outputs for inputs not encountered in training
- **Properties of Neural Networks**
  - Non Linearity -- Neurons are distributed throughout the network eg speech signal
  - Input – Output Mapping -- Reordering of previously trained examples and mapping Input – output using estimation statistics
  - Adaptivity -- Stability Plasticity Dilemma
  - Evidential Response -- Decision and Confidence , rejection of unwanted patterns
  - Contextual Information – Every neuron is affected by the activity of global neurons
  - Fault Tolerance -- hardware is fault tolerant, robust in nature , performance degrades in adverse conditions
  - VLSI Implementability -- well suited due to massive parallelism
  - Uniformity of Analysis and Design -- commonality, share and integrity
  - Neurobiological Analogy

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- **Human Brain**

The **human brain** contains about 10 billion nerve cells, or neurons. On average, each neuron is connected to other neurons through approximately 10,000 synapses.

	processing elements	element size	energy use	processing speed	style of computation	fault tolerant	learns	intelligent, conscious
	10^{14} synapses	10^{-6} m	30 W	100 Hz	parallel, distributed	yes	yes	usually
	10^{8} transistors	10^{-6} m	30 W (CPU)	10^{9} Hz	serial, centralized	no	a little	not (yet)

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Characteristics of BNN and ANN

Characteristics	Biological Neural Network	Artificial Neural Network
Speed	Processes information at a slower rate. Response time is measured in milliseconds.	Information is processed at a faster rate. The response time is measured in nanoseconds.
Processing	Massively parallel processing.	Serial processing.
Size & Complexity	An extremely intricate and dense network of linked neurons of the order of 1011 neurons and 1015 interconnections.	Size and complexity are reduced. It is incapable of performing sophisticated pattern recognition tasks.
Storage	An extremely intricate and dense network of linked neurons with 1015 interconnections, including neurons on the order of 1011.	The term "replaceable information storage" refers to the practice of replacing fresh data with old data.
Fault tolerance	The fact that information storage is flexible means that new information may be added by altering the connectivity strengths without deleting existing information.	Intolerant of faults. In the event of a system failure, corrupt data cannot be recovered.
Control Mechanism	There is no unique control mechanism outside of the computational task.	Controlling computer activity is handled by a control unit.

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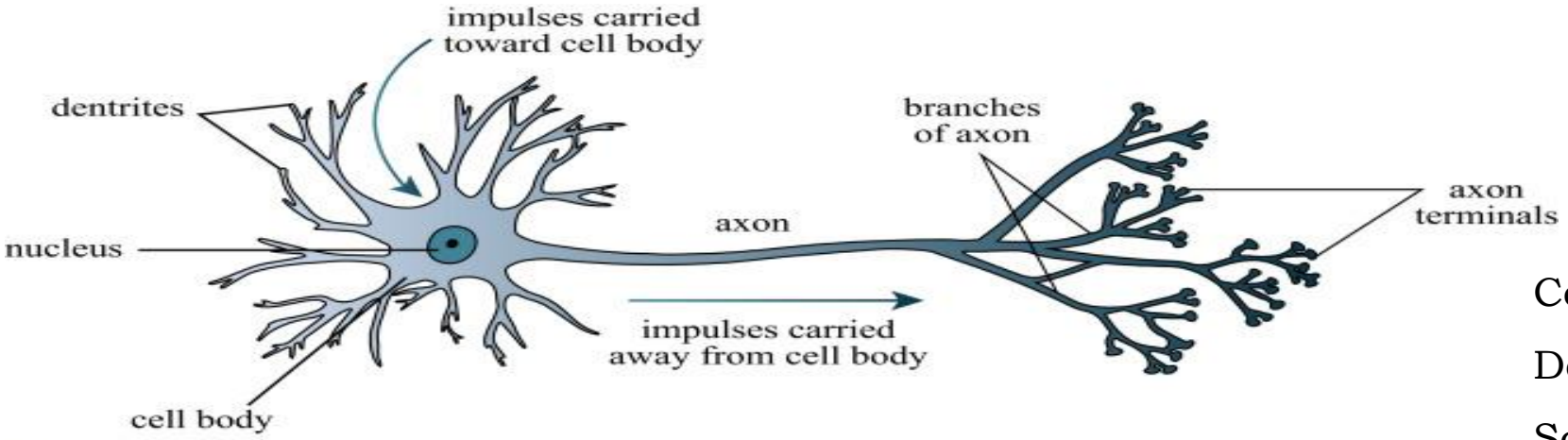
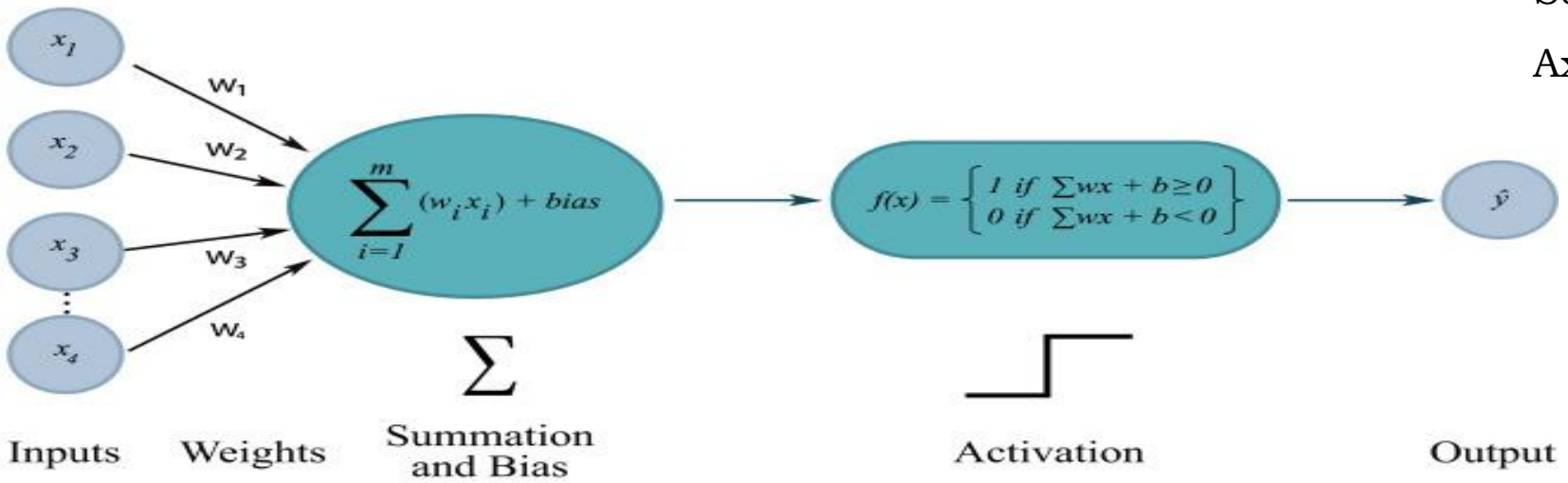


Cell – Neuron

Dendrites –Weight or interconnections

Soma – Net Input

Axon –Output



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Models of a Neuron

Synapses are the connecting junction between axon and dendrites. The majority of synapses send signals from the axon of a neuron to the dendrite of another neuron

Dendrites have fibers branching out from the soma in a bushy network around the nerve cell.

- Dendrites allow the cell to receive signals from connected neighboring neurons and each dendrite is able to perform multiplication by that dendrite's weight value

Axons are the single, long fibers extending from the main soma.

- The axon will branch and connect to other dendrites

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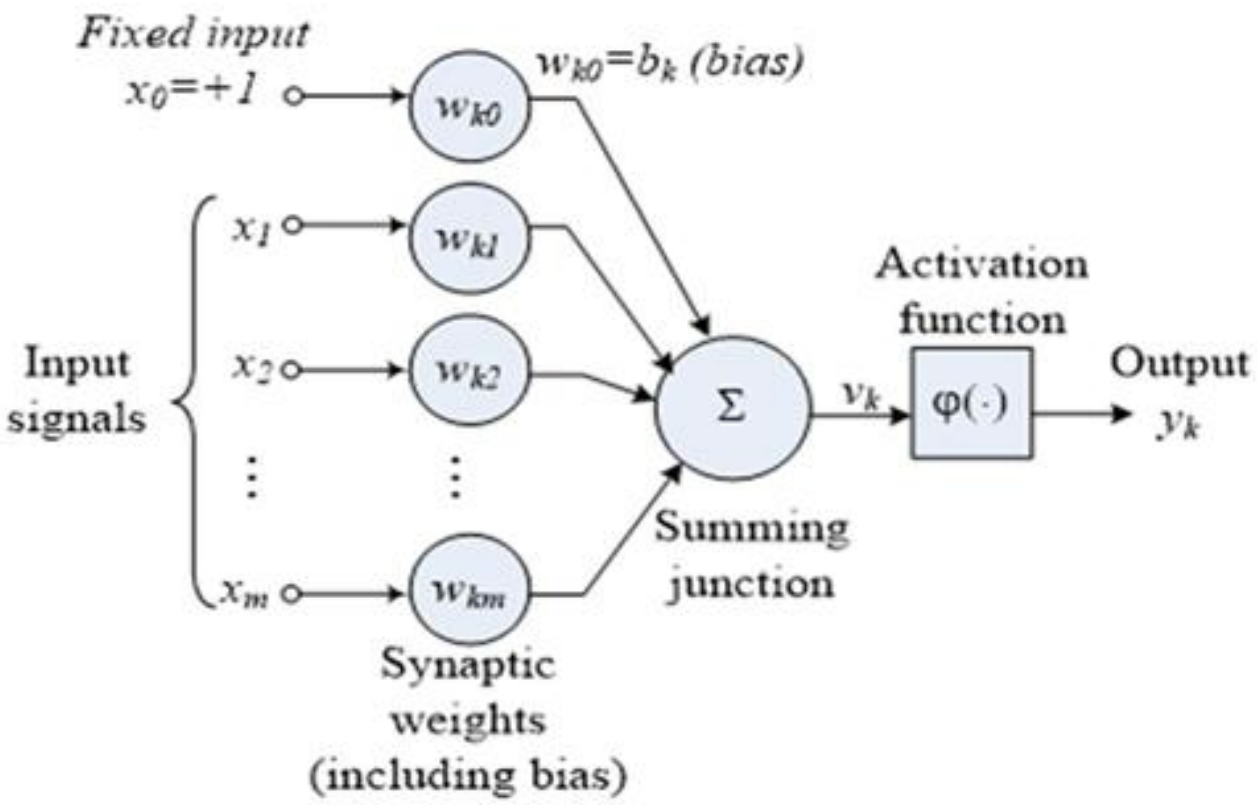
Non Linear Model of a Neuron



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Functional Parameters of a Neuron

Function parameter	Description
w	Vector of real-valued weights on the connections
w · x	Dot product ($\sum_{i=1}^{n} w_i x_i$)
n	Number of inputs to the perceptron
b	The bias term (input value does not affect its value; shifts decision boundary away from origin)

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Functional Parameters of a Neuron

Weight : Each Neuron is connected to other by means of communication link and each link is associated with weight.

- Weights contain information about the input signal that is used by the net to solve a problem.
- Represented in the form of a matrix and called as connection matrix.

Bias : Included in the network and has its impact in calculating the net input.

- Two types – Positive and Negative Bias
- Positive – helps to increase the net input
- Negative – Decreases the net input

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Functional Parameters of a Neuron

- Threshold : Is a set value based upon which the final output of the network is calculated.
  - Used in activation function
- Learning Rate : Denoted by ∞ . Used to control the amount of weight adjustment at each step of training
  - Ranges from 0 to 1
- Momentum Factor : Convergence is made faster if this is used.
  - Weights from one or more previous training patterns will be saved

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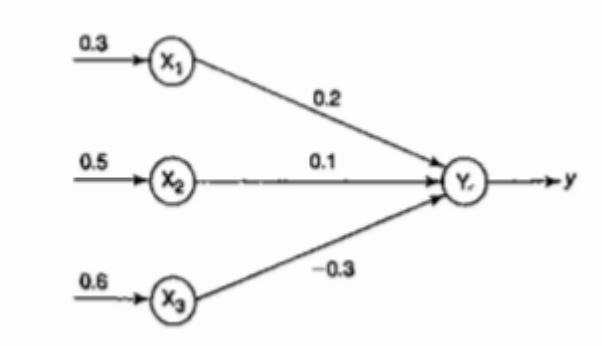
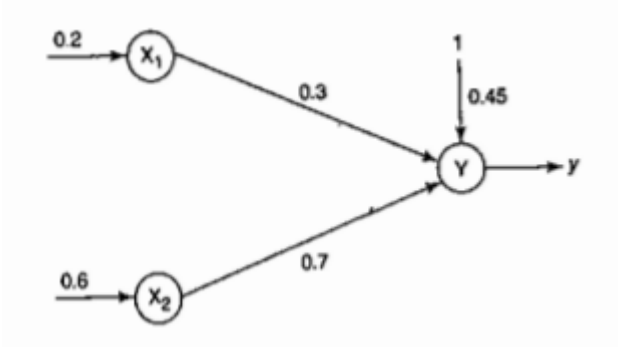
Problems

- For the given network, calculate the net input to the output neuron



- Calculate the net input (i) Without bias and (ii) when the bias is 0.45 and observe the difference and discuss



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Activation Function is a mathematical function used in artificial neural networks to determine whether a neuron should be activated or not.

- **Add Non-linearity:** Helps neural networks model complex data like images, speech, and text.
- **Control Output Range:** Maps input values into specific ranges (e.g., 0–1 or –1–1).
- **Enable Learning:** Allows gradient-based optimization during backpropagation.

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Type	Formula / Definition	Output Range	Key Points
Linear	Identity mapping	$(-\infty, \infty)$	Used in regression; not suitable for hidden layers
Sigmoid	Maps values between 0 and 1	(0, 1)	Used in binary classification; can cause vanishing gradient
Tanh	(-1, 1)	Zero-centered; smoother than sigmoid	
ReLU	0 for $x<0$, x for $x\geq 0$	$[0, \infty)$	Fast and popular; can cause "dying ReLU" problem
Leaky ReLU	$(-\infty, \infty)$	Allows small negative gradient; avoids dying ReLU	
Softmax	(0, 1), sum=1	Converts logits into probabilities for multi-class classification	

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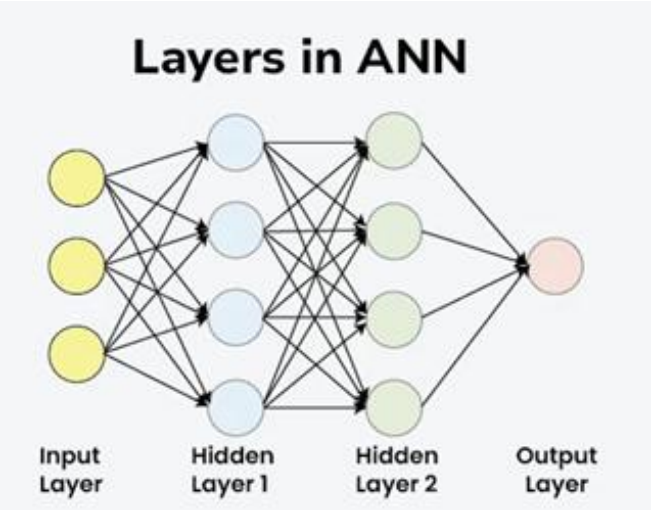
Layers and Connections

Input Layer

- Input layer is the first layer in an ANN and is responsible for receiving the raw input data.
- This layer's neurons correspond to the features in the input data.
- For example, in image processing, each neuron might represent a pixel value.
- The input layer doesn't perform any computations but passes the data to the next layer.
- Role: Receives raw data.
- Function: Passes data to the hidden layers.



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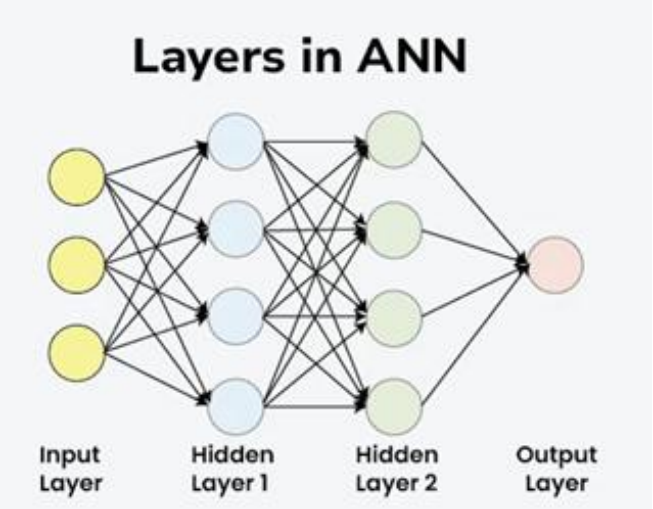
Layers and Connections

Hidden Layer

- Hidden Layers are the intermediate layers between the input and output layers.
- They perform most of the computations required by the network.
- Hidden layers can vary in number and size, depending on the complexity of the task.
- Each hidden layer applies a set of weights and biases to the input data, followed by an activation function to introduce non-linearity.



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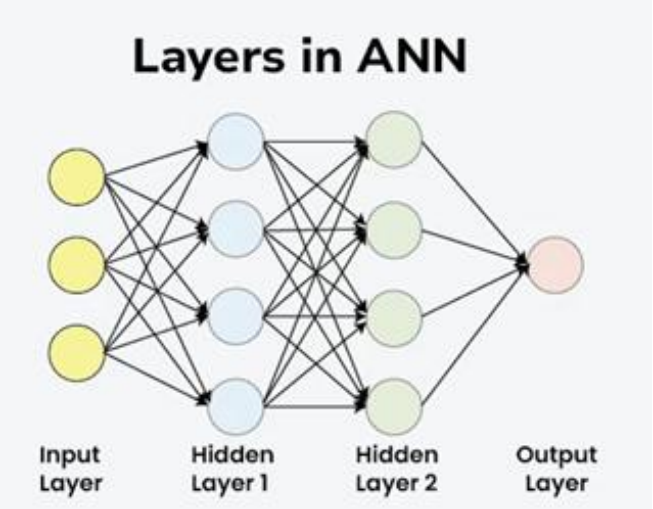
Layers and Connections

Output Layer

- Output Layer is the final layer in an ANN.
- It produces the output predictions.
- The number of neurons in this layer corresponds to the number of classes in a classification problem or the number of outputs in a regression problem.
- The activation function used in the output layer depends on the type of problem



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Layers and Connections

Types of Hidden Layer

Hidden Layer Type	Description / Function	Purpose	Common Applications
1. Dense (Fully Connected) Layer	Every neuron connects to all neurons in the previous and next layer.	Learns global patterns and relationships.	Feedforward Neural Networks (FNN), MLPs
2. Convolutional Layer (Conv Layer)	Applies filters (kernels) to extract local spatial features.	Detects edges, textures, and spatial hierarchies.	CNNs for image, video, and audio processing
3. Pooling Layer	Reduces spatial dimensions by summarizing nearby outputs (e.g., max, average).	Decreases computation, prevents overfitting.	CNNs (e.g., VGG, ResNet)
4. Recurrent Layer (RNN, LSTM, GRU)	Includes feedback connections allowing information to persist.	Captures temporal/sequential dependencies.	RNNs, LSTMs for NLP, time-series data
5. Normalization Layer	Normalizes inputs across mini-batches or features.	Stabilizes training, prevents gradient issues.	CNNs, RNNs, Transformers
6. Dropout Layer	Randomly disables a fraction of neurons during training.	Prevents overfitting, improves generalization.	DNNs, CNNs, RNNs
7. Flatten Layer	Converts multidimensional data (e.g., 2D images) into 1D vectors.	Prepares data for fully connected layers.	Transition between CNNs and dense layers

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Layers and Connections

Connection Type	Description	Example / Application
Feedforward Connections	Information moves only forward (input → hidden → output).	Standard Neural Networks
Recurrent Connections	Output of neurons feeds back into the same or previous layers.	RNNs, LSTMs
Skip / Residual Connections	Directly connect non-adjacent layers to improve gradient flow.	ResNet, Transformers
Convolutional Connections	Local connections using shared weights (filters).	CNNs
Fully Connected (Dense) Connections	Every neuron in one layer connects to every neuron in the next.	MLPs (Multilayer Perceptrons)

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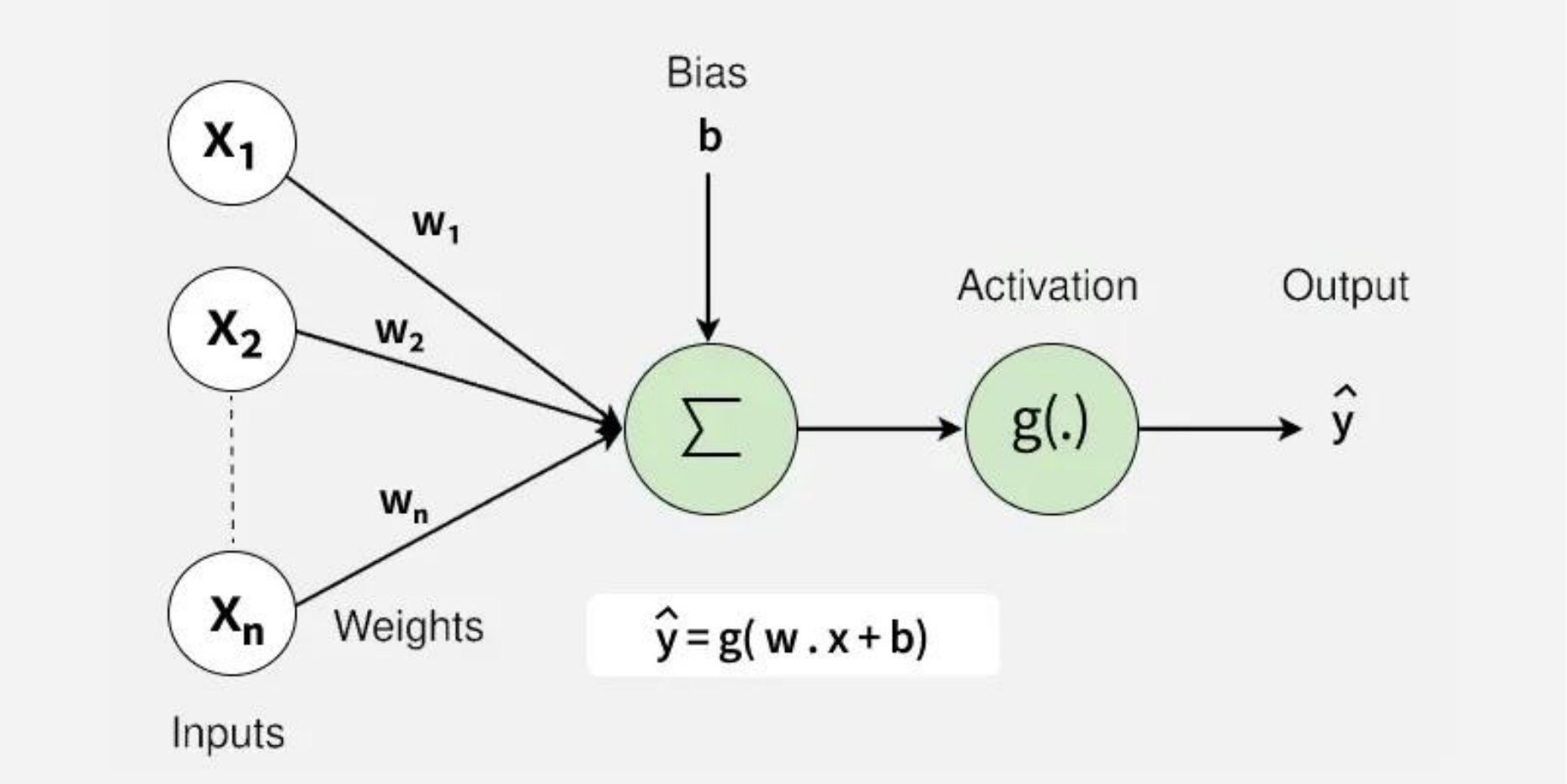
Types of Neural Network Architectures

Basic Structure



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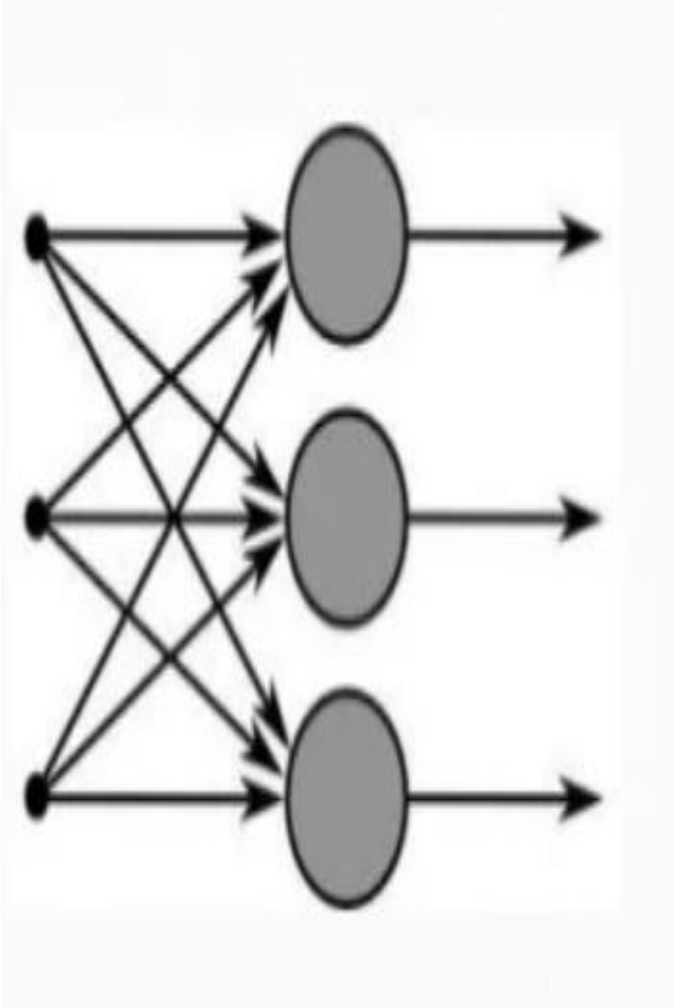
Network Architeures

- Single Layer Feedforward network
  - Consists of input layer of source nodes that projects to output layer of computation node
  - Strictly feedforward or acylic type
  - Single output layer of computation nodes
- Layer is formed by taking a processing element and combining it with other processing elements
- Interconnections lead to the formation of various network architectures



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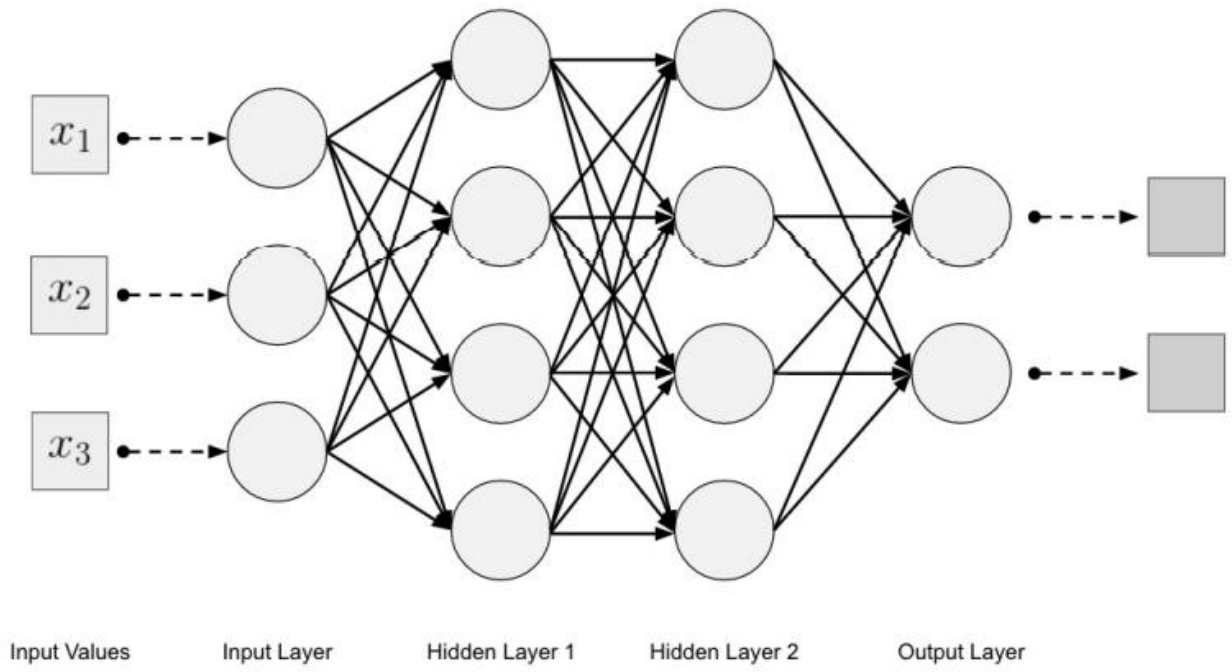
Network Architeures

- Multi Layer Feedforward network



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Perceptron

- The Perceptron is the simplest type of artificial neuron, introduced by Frank Rosenblatt(1958).
- It is a binary classifier that decides whether an input belongs to one class or another based on a linear combination of inputs and weights.
- Perceptron consists of a single layer of input nodes that are fully connected to a layer of output nodes.
- This foundational model has played a crucial role in the development of more advanced neural networks and machine learning algorithms.

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Types of Perceptron

Single-Layer Perceptron is a type of perceptron is limited to learning linearly separable patterns.

- It is effective for tasks where the data can be divided into distinct categories through a straight line.
- While powerful in its simplicity, it struggles with more complex problems where the relationship between inputs and outputs is non-linear.

Multi-Layer Perceptron possess enhanced processing capabilities as they consist of two or more layers, adept at handling more complex patterns and relationships within the data.

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Components of Perceptron

Input Features: The perceptron takes multiple input features, each representing a characteristic of the input data.

Weights: Each input feature is assigned a weight that determines its influence on the output.

These weights are adjusted during training to find the optimal values.

Summation Function: The perceptron calculates the weighted sum of its inputs, combining them with their respective weights.

Activation Function: The weighted sum is passed through the Heaviside step function, comparing it to a threshold to produce a binary output (0 or 1).

Output: The final output is determined by the activation function, often used for binary classification tasks.

Bias: The bias term helps the perceptron make adjustments independent of the input, improving its flexibility in learning.

Learning Algorithm: The perceptron adjusts its weights and bias using a learning algorithm, such as the Perceptron Learning Rule, to minimize prediction errors

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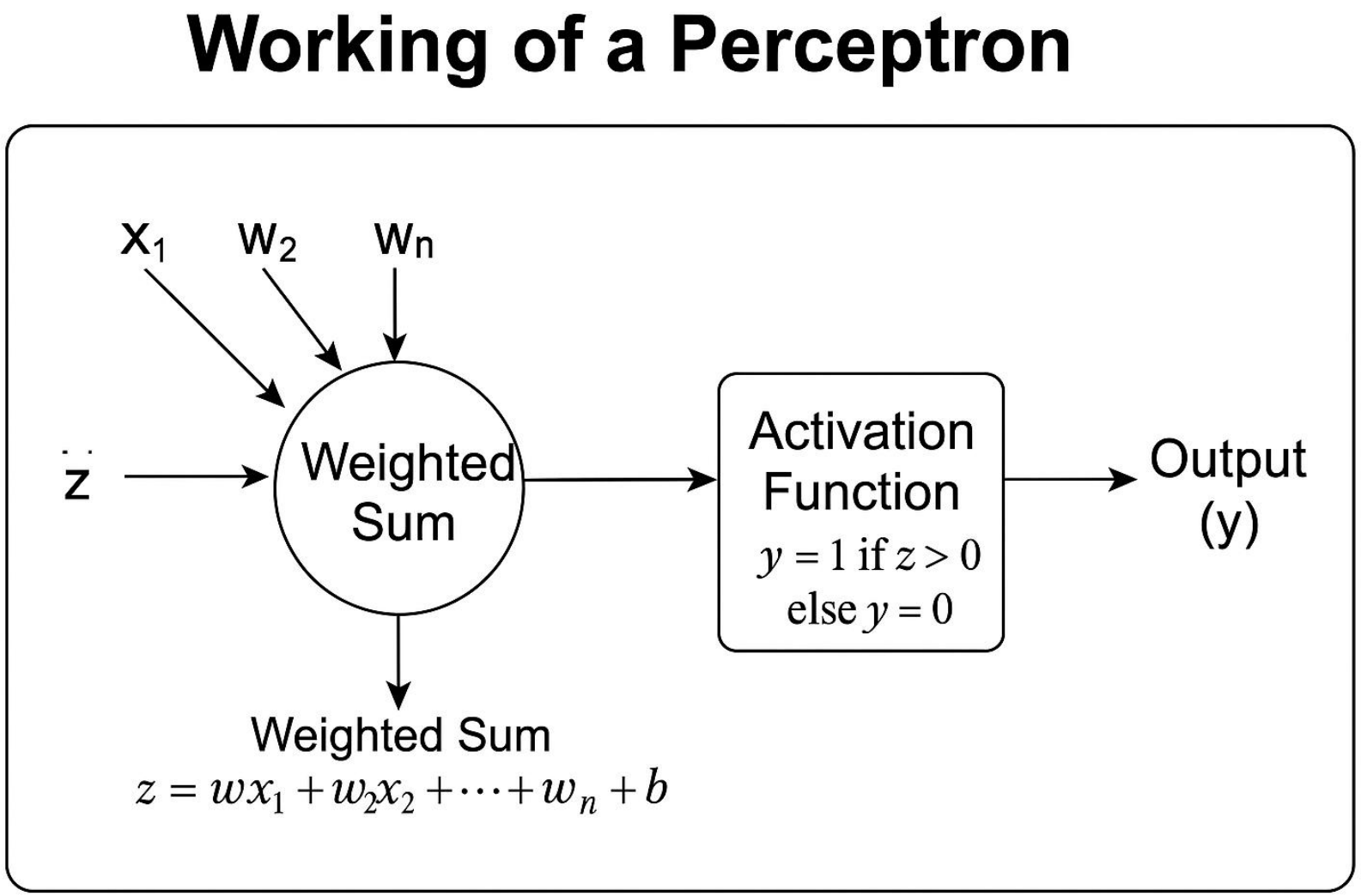
Working of Perceptron



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Working of Perceptron

Step	Process	Explanation
1. Input Signal	(x_1, x_2, x_3, ..., x_n)	Each input represents a feature (e.g., pixel value, sensor reading, etc.).
2. Weight Assignment	(w_1, w_2, w_3, ..., w_n)	Each input is assigned a weight that represents its importance.
3. Weighted Sum Calculation	(z = w_1x_1 + w_2x_2 + ... + w_nx_n + b)	The perceptron computes the weighted sum of all inputs and adds a bias term (b).
4. Activation Function (Decision Making)	(y = f(z))	The result is passed through an activation (usually a step) function to decide the output.
5. Output Generation	(y = 1) if (z > 0), else (y = 0)	Produces a binary output: 1 (activated) or 0 (not activated).